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Wearable sensors for ECG measurement: a review

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Abstract

Purpose – Recent developments in wearable technologies have paved the way for continuous monitoring of the electrocardiogram (ECG) signal, without the need for any laboratory settings. A number of wearable sensors ranging from wet electrode sensors to dry sensors, textile-based sensors, knitted integrated sensors (KIS) and planar fashionable circuit boards are used in ECG measurement. The purpose of this study is to carry out a comparative study of the different sensors used for ECG measurements. The current challenges faced in developing wearable ECG sensors are also reviewed.

Design/methodology/approach – This study carries out a comparative analysis of different wearable ECG sensors on the basis of four important aspects: materials and methods used to develop the sensors, working principle, implementation and performance. Each of the aspects has been reviewed with regard to the main types of wearable ECG sensors available.

Findings – A comparative study of the sensors helps understand the differences in their operating principles. While some sensors may have a higher efficiency, the others might ensure more user comfort. It is important to strike the right balance between the various aspects influencing the sensor performance.

Originality/value – Wearable ECG sensors have revolutionized the world of ambulatory ECG monitoring and helped in the treatment of many cardiovascular diseases. A comparative study of the available technologies will help both doctors and researchers gain an understanding of the shortcomings in the existing systems.

Keywords Capacitive coupling electrodes, Electrocardiogram (ECG), Knitted integrated sensors, Planar fashionable circuit boards, Wearable sensor, Wet electrodes

Paper type General review

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1. Introduction

Over the past few decades, there has been an exponential increase in the general health awareness of people. Medical advancements have revolutionized the treatment of a wide range of critical diseases. However, most of these technologies are not portable and require medically trained experts for their implementation. The need to continuously track patients' physical activities and physiological parameters in the external environment led to the development of wearable health monitoring systems (Pantelopoulos and Bourbakis, 2010). They promise to change the face of medical check-ups by replacing one-off tests with continuous monitoring of vital signals (Bloss, 2015). Glaucoma, heart attack and diabetes are few of the many life-threatening conditions that can be detected and efficiently treated using wearable technology. Figure 1 depicts the range of wearable systems that are used to measure different physical parameters of the body.

Wearable systems are prominently used in the treatment of cardiovascular diseases (CVD). These include numerous heart-related complications such as cardiac arrest, arrhythmia, congestive heart failure, coronary artery disease, etc. According to an update by the American Heart Association, in 2011, the age-standardized death rate attributable to all CVDs was as high as 229.6 per 100,000 in the USA alone (Lloyd-Jones *et al.*, 2010). Delayed detection of CVDs is a major reason for the alarmingly high death rates.

An electrocardiogram (ECG) is a diagnostic tool to measure the electrical signals that control various activities of the heart. A resting ECG merely captures the heart's functioning at a given moment and does not account for irregularities that may occur at a later point of time (Oresko *et al.*, 2010). Continuous monitoring of the ECG promises early detection and treatment of CVDs. This has resulted in the development of a number of wearable applications for continuous ECG monitoring.

The traditional ECG systems were based on wet Ag/AgCl electrodes directly attached to the patient's skin using an electrolyte gel. These sensors were highly inconvenient for long-term applications as they led to skin rashes and allergies on prolonged use (Searle and Kirkup, 2000). Thus, dry electrodes

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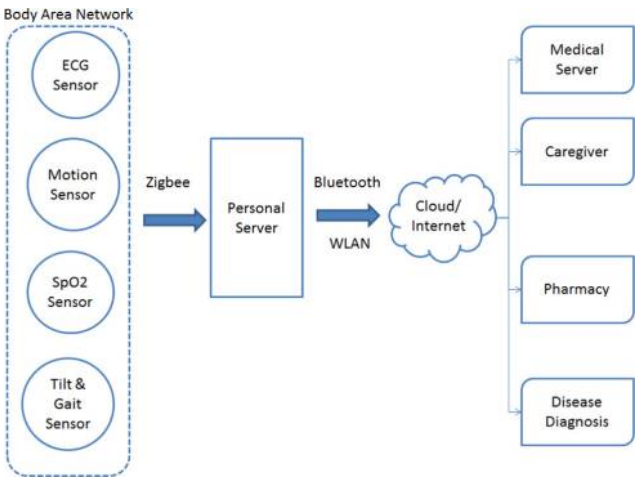
Figure 1 The range of wearable systems used to measure different physiological parameters



were developed, with the view of omitting the need for electrolyte gel. While being more comfortable than wet electrodes, they compromised the patient's safety due to direct electrical contact between the skin and the electrode. The need for comfort and safety of the patients led to the development of capacitively coupled electrodes. These are non-contact electrodes that work on the principle of capacitive coupling of charges between the patient's skin and the electrode. These systems were initially developed to measure a number of physiological parameters such as gait recognition (Baldwin *et al.*, 2015) and respiration monitoring (Merritt *et al.*, 2009). Currently, they have evolved into an established sensing technique for ECG. Table I presents a detailed comparison of the three main types of ECG electrodes: wet, dry and capacitively coupled electrodes.

Corresponding to the growth of wearable sensors, there has been an equal rise in the use of SMART fabrics, where a conductive yarn is used to develop sensing properties (Lymberis and Paradiso, 2008). These systems are known as knitted integrated sensors (KISs). The electronic components are woven into the conductive textiles improving comfort levels of the sensor. The most recent advancements in ECG sensors are the use of planar fashionable circuit boards (PFCB) that comprise the entire circuit being printed on the fabric using a silver paste.

The paper aims to draw a comparative study of the different types of wearable ECG sensors such as wet and dry electrodes, capacitive coupling sensors, KIS and PFCBs. Section 2 discusses the various materials used and the methodologies adopted for the fabrication of different sensors. The material requirements vary according to the type and sensing principle of various sensors. Section 3 reviews the basic working principle of each sensor and how these sensors differ from each other. Section 4 explains the signal conditioning circuits required for the sensors, and Section 5 compares the performance of each sensor. Finally, Section 6 discusses the future scope and challenges faced in wearable ECG sensing.

2. Materials and methods

Wearable technologies aim to make ECG sensors an intrinsic part of a person's everyday life. The materials and methods used in fabricating an ECG sensor are an important factor determining the comfort level offered by a sensor.

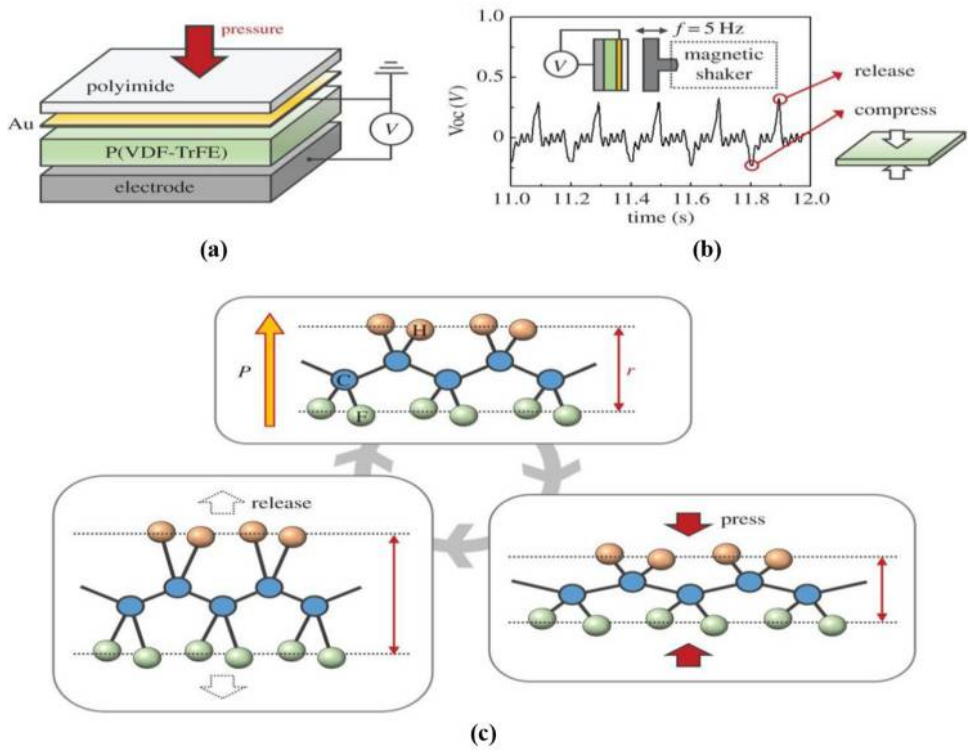
To ensure the user's convenience, the textile chosen for the sensor should be such that it is soft, does not reduce the skin's breathability and is aesthetic (Fuhrhop *et al.*, 2009). A majority of the sensors are made using electroactive polymeric materials. The flexibility, lightness and intrinsic sensing properties offered by these materials make them suitable for fabricating wearable ECG electrodes (Finlay *et al.*, 2008). Figure 2 represents the working of an electroactive polymer.

The most common method of developing ECG electrodes involves fabricating metal electrodes onto a polymer substrate. Most of these systems use the elastomer poly (dimethyl siloxane) [PDMS] as an electroactive polymer substrate. It is known for its biocompatibility and the ease with which micro patterns can be moulded onto its surface (Finlay *et al.*, 2008). The nickel (Ni)–phosphorus (P)-plated polyester fibre is another commonly used polymer for wearable sensors. When compared with the wet Ag/AgCl-based electrode, a small square piece of the printed fabric gave results comparable to that of the standard electrode (Haghdoost *et al.*, 2015). Although these electrodes produced more noise than the wet electrodes, they have the advantage of eliminating the need of an electrolyte gel. However, it is also important to maintain

Table I Comparison of wet, dry and capacitively coupled electrodes

Electrode Type	Ag/AgCl electrodes (Wet electrode)	Dry electrodes	Capacitively coupled electrodes
Description	It uses a solid/liquid gel as an electrolyte to establish contact with the skin	It comprises metal disc/ conductive plastics that are placed on the skin. The sweat acts as an electrolyte	The electrode is separated from the skin by a high dielectric material (such as cloth)
Principle of operation	Charge transfer via a resistive path between the skin and the electrode	Charge transferred via a resistive path between the skin and the electrode	Charge transferred via capacitive coupling between skin and electrode
Advantages	Good signal quality Intensive care No local electronics needed	Comfortable, reliable Less skin irritation Permanent monitoring	Good signal quality No electrode artefacts No skin irritation
Disadvantages	Requires expert assistance for use Frequent exchange Skin irritation	Low signal quality High electrode–skin and input impedance	Local electronics required Shielding is important

Figure 2 (a) Metal electrode fabricated on an electroactive polymer, (b) the variation in the electric potential when pressure is applied and then released from the polymer surface and (c) the changes in the molecular structure of the polymer with the variation of the pressure applied



Source: Wang *et al.* (2016)

continuous contact between the skin and the electrode. Baek *et al.* (2008) developed a PDMS-based dry electrode, capable of changing its shape to maintain constant contact between the skin and the electrode. When the sensor was subjected to a peel-off test, the metal pattern remained successfully adhered to the PDMS layer.

The KISs are based on piezoresistive materials known as SMART textiles or e-textiles. They are capable of sensing various physical parameters and elicit appropriate responses. To convert physical signals into electrical ones, smart textiles are usually made from electro-conductive polymers (Van Langenhove and Hertleer, 2004). They are capable of producing a wide range of voltages ranging from a few millivolts to a few hundreds of volts (Edmison *et al.*, 2002). They also provide an array of form factors including coaxial cable, paints, film and threads. This coupled with their low power requirements and the ability to recognize almost any physical stimuli makes them a perfect fit for KIS.

The WEALTHY system incorporating the use of piezoelectric-based sensing was the foremost KIS to be developed (Paradiso *et al.*, 2005). It comprised strain fabric sensors and metal yarns for the electrodes. An important factor affecting the piezoresistivity of the sensor is the number of loops present in the same course of the yarn (Paradiso *et al.*, 2005), as it affects the length of the path taken by the charges. Thus, the resistivity of sensors with the same surface area changes corresponding to a change in the courses and loops of the yarn. Figure 3 represents the stages involved developing a KIS.

The system on textile approach is also used for these sensors. The fabric substrate comprises a number of insulated metal yarn interconnects forming a grid called the "textile via". It serves as the fundamental building block for the sensor and determines the electrical interconnections present within the fabric (Locher and Troster, 2007).The electronic components are then integrated on to the substrate using interposer pads. Elasticity and the extent of adherence are the two key factors affecting the performance of the KIS and depend upon the type of knitting used in the sensor. Seamless knitting systems, where the electrodes and sensors are knitted together in one single manufacturing step, generally have improved elasticity with good comfort levels (Paradiso and Caldani, 2010).

While the KISs use conductive yarn as the substrate, the PFCBs use any casual textile such as a denim patch as the substrate and the electrodes are deposited on to the fabric (Kim *et al.*, 2008). The commonly used methods of etching electrodes on to the fabric surface are gold sputtering technique and silk screening of conducting epoxy. Passive elements such as resistors, capacitors and inductors are also developed on the planar circuit as per the requirements of the system. Figure 4 represents the silk printing and sputtering methods of PFCB fabrication.

3. Working principle

The skin electrode interface is the most important part of the wearable ECG sensor. This interface mainly comprises multiple conductive and capacitive coupling layers between the

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Figure 3 The various steps involved in the making of a knitted integrated sensor

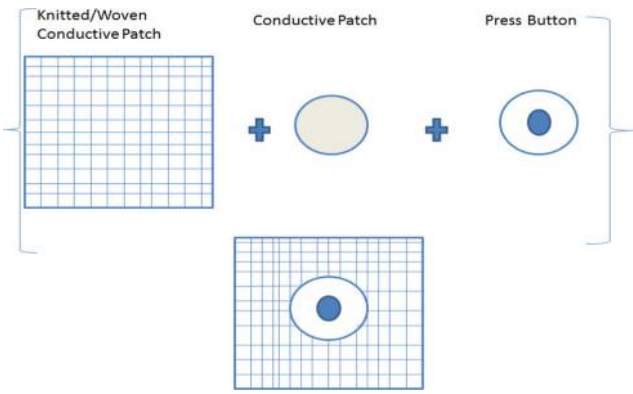
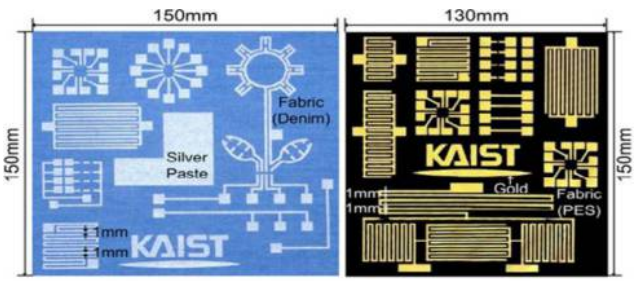


Figure 4 Fabrication of PFCB by (a) silk printing and (b) sputtering



Source: Kim *et al.* (2009)

skin and the electrode. These layers are complemented by a number of serially connected parallel resistor–capacitor (RC) networks. The type and construction of each sensor varies such that different RC networks are predominant in different sensors and this determines their basic operating principle (Chi *et al.*, 2010).

The conventional wet electrodes work on the principle of electrolytic half-cell potentials (Chi *et al.*, 2010). These potentials are coupled along with series resistances between the skin and the electrode. The resistors in turn are a part of the Nernst path between the skin and the electrolyte (Nemati *et al.*, 2012). Thus, the wet electrode is a contact sensor that uses the resistance between the layers of the skin and the electrode to measure the electrical activities of the heart. The electrolyte gel acts as a medium of contact and helps transfer charges between the two layers (Fensli *et al.*, 2005).

However, the electrolyte gel used is highly inconvenient for long-term applications. To overcome this problem, researchers have developed dry electrodes that are contact electrodes without the need for any hydro gel. When the electrode is placed on the skin, sweat accumulates around the surface due to perspiration. The sweat being primarily composed of sodium chloride resembles the electrolyte solution chemically and hence acts as a substitute for the same (Meziane *et al.*, 2013). Thus, dry electrodes merely mimic the wet electrode in their working after successfully eliminating the need for a gel or any other sort of skin preparation.

The capacitive coupling sensors are non-contact sensors that do not require any direct contact with the patient's skin. A thin dielectric material is placed between the metal electrode and the skin. Charge transfer from the skin to the electrode takes place by means of capacitive coupling at the skin–dielectric and dielectric–electrode interface (Nemati *et al.*, 2012).The most commonly used dielectric materials are textiles. This has led to the emergence of an array of textile-based sensors for ECG and other wearable technologies

The constant attempt to make the ECG sensors comfortable and patient friendly led to the development of KIS. They are based on the concept: "fabric is the computer" (Park *et al.*, 2002). The different electronic devices required for the sensor circuit are seamlessly knitted into the fabric with the help of different types of conductive yarn. Almost all electronic circuits can be woven into the fabric with alternate layers of conductive and non-conductive yarn (Post *et al.*, 2000). Once woven into the fabric, these sensors work on the same principle as that of other wearable ECG sensors. The PFCBs are also similar in their principle of operation. They resemble a conventional PCB (printed circuit board), with the exception that the circuit is printed directly on the fabric instead of plastic (Kim *et al.*, 2010). Figure 5 draws a comparison of the working principle of different sensors.

4. Implementation

Any ECG sensor comprises two important parts: the electrode and the signal conditioning circuit. The different types of electrodes and their working principles have been discussed in the preceding section. This section throws light on the signal conditioning circuit which helps ensure that the signal picked up by the electrodes is processed and transmitted to the base station without any distortion. Figure 6 gives a block diagram representation of ECG signal conditioning circuit.

The primary aim of any signal conditioning circuit in wearable ECG sensors is to compensate for the skin electrode contact impedance. The skin's outermost surface being a dry dielectric surface serves as a barrier for any transfer of charges between the skin and the electrode (Gruetzmann *et al.*, 2007).

In wet electrodes, the effect of the skin electrode impedance is significantly reduced by the electrolyte gel, which serves as a conducting layer (Spach *et al.*, 1966). However, in dry and non-contact sensors, this impedance value is higher owing to the absence of an electrolyte gel. The lack of a wet conductive medium is generally compensated for using high-input impedance buffers (Chi *et al.*, 2009). These buffers help reduce the skin electrode impedances to a negligible value. The impedance value also varies at different parts of the body. Any reliable ECG sensor should be able to accommodate impedances as high as 100 kΩ (Spach *et al.*, 1966).

Most signal conditioning circuits comprise a front-end amplifier and a differential amplifier. The front-end amplifier is designed to have a high-input impedance along with low noise levels The most commonly used front-end amplifier is the INA116, which is prominent for its ultrahigh = input impedance values and extremely low current noise (Chi and Cauwenberghs, 2010). The second amplifier in the signal conditioning circuit which is a differential amplifier is generally realized by an FET amplifier such as LTC6078 as shown in Chi *et al.* (2009). This amplifier provides a direct current (DC) path using an input

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Figure 5 Comparison of working principle of different types of wearable ECG sensors

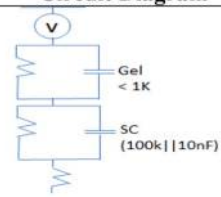
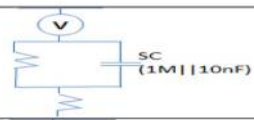
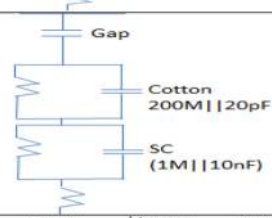
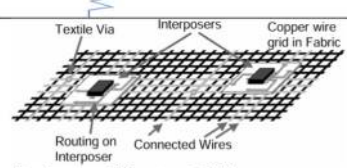
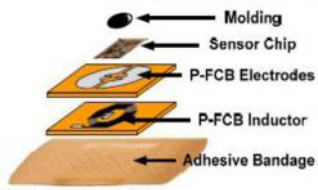
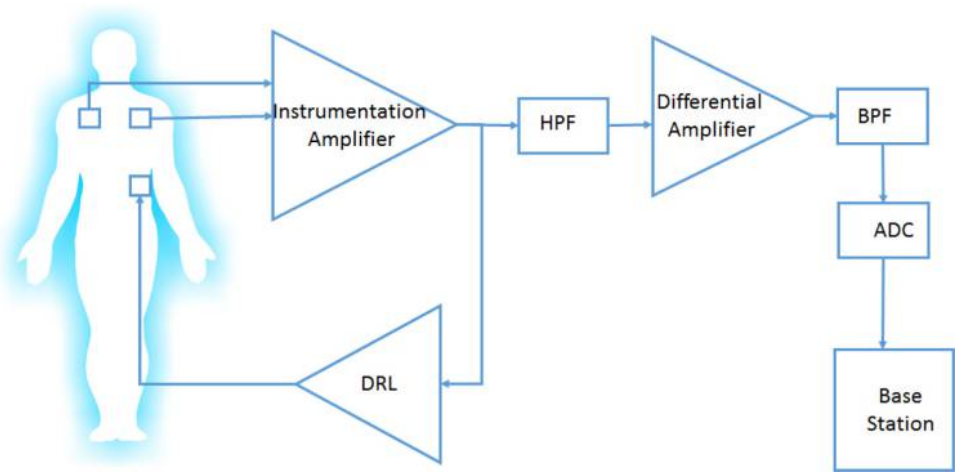
Sensor	Principle of Operation	Circuit Diagram
Wet Electrode Sensor	Charges are transferred via a resistive path between the skin and the electrode Works on the principle of electrolytic half cells	
Dry Contact Sensor	Similar to wet electrode. Electrolyte get is replaced by sweat from the skin	
Capacitive Coupling Sensor	Charge transferred by capacitive coupling between skin and electrode via a dielectric material.	
Knitted Integrated Sensor	The electronic components are woven into the fabric	 (Locher and Troster, 2007)
P-FCB	The electronic circuit is printed directly on the fabric.	 (Yoo *et al.*, 2009b)

Figure 6 A block diagram of the signal conditioning circuit used in wearable ECG sensors



biasing circuit. Table II provides a comparison of the commonly used amplifier ICs.

The wearable sensors are mainly affected by the external noise and the common mode voltage error. The environmental noise is countered using an active shield that guards the capacitance from any externally coupled noise. Common mode voltage rejection is achieved using a driven right leg circuit (DRL) (Spinelli *et al.*, 1999). It comprises an op amp which

Table II List of commonly used amplifier ICs and their specifications

Amplifier IC	Supply voltage Min.(V)	Supply voltage Max.(V)	Maximum output current (in mA)	Features
LMC6001	−0.3	+16.0	±30	Low power, low noise
INA 106	±5	±18	±20	Precision gain, high CMRR
INA116	±4.5	±18	±1.4	Low input bias current, buffered guard pins
LT6010	−20	+20	±10	Low noise, rail-to-rail output swing
LTC6078	−3	+3	Indefinite	Micro power, rail-to-rail inputs and outputs
LMP7702	−0.3	+0.3	±42	Rail-to-rail outputs, precision gain
LMC6081	−0.3	+0.3	±30	Ultra-low bias current, improved latch-up immunity

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gives negative feedback to the reference electrode and thus helps in rejecting the common-mode noise. The DRL can also be replaced by a time-averaged common-mode voltage line, aggregated over all the electrodes (Chi *et al.*, 2009).

The implementation procedure differs for the KIS as the components used are directly integrated on to the fabric. The fundamental building blocks of fabric-based circuits are the transmission lines. These lines contribute towards the maximum signal frequency (bandwidth) and impedance of the sensor (Locher and Troster, 2007). The PFCBs are directly screen printed on to the fabric. Once the circuit has been designed, the silver paste is printed onto the fabric (Yoo *et al.*, 2009a). The diameter of the electrodes and distance between them are optimized as per the design considerations of the system.

5. Performance

The power consumption, size, sensor gain and noise are some of the important parameters that help determine the efficiency of a given ECG system. Ideally, a wearable ECG system should be an extremely light, ultra-low power system with a long battery life to avoid repeated replacement of the battery.

The sensor gain of the system is a variable that depends upon the distance between the electrode and the skin. The gain varies as a function of the skin electrode impedance which in turn is dependent on the distance between the skin surface and the electrode (Chi *et al.*, 2009). There is a general decrease in the gain with an increase in the distance. The gain is 869 at a distance of 0.2 mm which falls to 539 as distance is increased to 1.6 mm and further drops to 391 at a distance of 3.2 mm (Loriga *et al.*, 2006). To counter the distance-dependent nature of the gain, an active shield is used instead of a passive shield as it is capable of working over a wide range of distances.

The noise level of the circuit is another important performance parameter of an ECG sensor. It is mainly dependent on the input leakage current of the amplifier and directly proportional to the value of the current (Chi *et al.*, 2009). The referred-to-input noise for sensor distance of 0.2 mm is approximately 0.66 μVrms. This noise gets integrated by the capacitance at the amplifier input to a noise level of 1 μVrms. This is countered by the guard pins of the INA116 amplifier IC. Further, thermal noise at the input resistance leads to a measured noise level of 1.88 μVrms (Loriga *et al.*, 2006). The noise generally increases with an increase in the sensor distance.

The general overall sensor performance is better for dry or non-contact electrodes in comparison to the wet gel-based electrodes. This is mainly because of the skin potential variation (SPV), which develops when the skin shrinks or stretches under the pressure of the wet electrodes (Pei *et al.*, 2017). The SPV is largely non-existent in non-contact sensor. Moreover, the dry sensors can be used continuously for periods as long as seven days without causing any discomfort to the user (Baek *et al.*, 2008). The wet electrodes, on the contrary, can be used continuously only for a few hours as they generally lead to skin rashes and itching on prolonged use.

In textile-based sensors, the increase in impedance after subsequent washing is an important factor of consideration. Kaappa *et al.* (2017) found that metal electrodes show a lesser increase in impedance than the softer polymer-based electrodes. However, the latter is more convenient for patient use. It is important to strike the right balance between the various factors influencing a sensor's performance.

6. Conclusion

Wearable ECG systems promise to revolutionize medical treatments for various cardiovascular diseases (CVD). Chronically ill patients, children and old people stand to benefit the most from these systems. The older population are more prone to heart-related complications. Continuous monitoring of their ECG will help take preventive action at an early stage. These systems will help develop high-efficiency ambulatory ECG systems that will overcome the constraints of resting ECG systems.

Despite their convenience of use and continuous monitoring, the wearable ECG systems are rigged by an array of problems that severely hinder their performance. One major issue is the non-uniform nature of the skin electrode impedance. This impedance if not buffered tends to distort the ECG data greatly. However, its value varies at different points of the body and also at different times, thereby making it difficult to converge on a single skin electrode impedance value. Another major issue with regard to ECG systems is motion artefact. This refers to the noise in the ECG signal that results from the relative movement between the skin and the electrode. This problem is especially high in dry and non-contact sensors. Although tensile textiles are used to reduce the movement of the electrodes, they can never completely eliminate the issue of

motion artefact. Another important aspect of all wearable technologies is the secure communication of signals from the sensor to the base station. A stable system with a secured transmission protocol is of utmost importance while considering wearable sensors (Li, 2010).

The wearable ECG systems have come a very long way from the wet electrode-based Holter monitoring systems to dry non-contact sensors and small unobtrusive circuits that can be woven or directly printed on to the fabric. The constant need to improve medical care for CVDs promises an ever-growing advancement in wearable ECG systems.

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